Yanwei Sun

PhD Candidate in Analytics & Operations, Imperial Business School
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Education

Imperial Business School, Imperial College London
Ph.D. in Analytics & Operations — 2022 – 2027 (expected)
Advisors: Prof. Jiahua Wu and Prof. Chiwei Yan
Master of Research in Business (Analytics & Operations) — 2022 – 2023

Beijing Institute of Technology
B.S. in Industrial Engineering (summa cum laude) — 2017 – 2021

Experience

University of California, Berkeley — September 2024 – August 2025
Visiting Scholar, hosted by Prof. Chiwei Yan

University of Washington, Seattle — July 2021 – August 2022
Research Affiliate (remote), working with Prof. Chiwei Yan

Research Interest

Broadly, I study **marketplace design** in settings where **participants are self-interested and information is dispersed**. My recent research focuses primarily on **markets for AI and markets with AI**.

- In markets for AI, where AI itself is the product, I study how AI models and services should be priced and contracted.
- In markets with AI, I investigate the new possibilities that AI brings to markets: AI agents can themselves be decision-makers, and AI-driven predictions can reshape market structures.

I have also worked on **information design and pricing in platform operations**, especially the interplay between information and pricing, and **dynamic service operations**, including three-sided marketplaces (e.g., food delivery systems) and on-off systems with self-interested agents.

Journal Papers (including minor revisions)

$*$: equal contribution; (α–β): alphabetical author order

1. (α–β) Zhe Liu, Yiwen Shen, *Yanwei Sun*. Operating Three-sided Marketplace: Pricing and Spatial Staffing in Food Delivery Systems. ***Minor Revision at Management Science***.
   - 🏆 First Place, INFORMS Service Science Best Cluster Paper Award, 2024.
   - 🏆 Second Place, CSAMSE Best Paper Award, 2024.
2. Fupeng Sun*, *Yanwei Sun**, Chiwei Yan, Li Jin. Restricting Entries to All-Pay Contests. ***International Journal of Game Theory***, 55, Article 7, 2026.

3. Chiwei Yan, James Johndrow, Dawn Woodard, *Yanwei Sun*. Efficiency of ETA Prediction. ***SIAM Journal on Mathematics of Data Science***, 6(2): 227–253, 2024.

Refereed Conference Proceedings

1. (α–β) Zhicheng Du, *Yanwei Sun*, Zihe Wang, Jiahua Wu. Prices as Signals: Joint Design of Information and Pricing. *22nd Conference on Web and Internet Economics (**WINE'26**)*, 2026.
2. (α–β) *Yanwei Sun*, Jiahua Wu, Jibang Wu. Beyond One Shot: Contracting for Sequential Trials. *Proceedings of the ACM Conference on Economics and Computation (**EC'26**)*, 2026.
3. (α–β) Niklas Häusle, Daniel Quigley, Casey Rothschild, *Yanwei Sun*, André Veiga. Optimal Privacy with Coarse Labels in Screening Markets. *Proceedings of the ACM Conference on Economics and Computation (**EC'26**)*, 2026.
4. *Yanwei Sun**, Fupeng Sun*, Chiwei Yan, Jiahua Wu. The Role of Prescreening in Auctions with Predictions. *Proceedings of the 21st Conference on Web and Internet Economics (**WINE'25**)*, 2025.
5. *Yanwei Sun**, Niloofar Zamani Foroushani*, Zhe Liu, Jiahua Wu. Information Design and Pricing in Lead-Selling Platforms with Supply Competition. *Proceedings of the ACM Conference on Economics and Computation (**EC'24**)*, 2024.
6. Fupeng Sun*, *Yanwei Sun**, Chiwei Yan, Li Jin. Restricting Entries to All-Pay Contests. *Proceedings of the ACM Conference on Economics and Computation (**EC'24**)*, 2024.

Working Papers

1. **[Job Market Paper]** (α–β) Ilan Lobel, *Yanwei Sun*, Jiahua Wu. Pricing AI as a Service. ***under review at Management Science***.
   - 🏆 Finalist (winner to be announced), Revenue Management and Pricing Jeff McGill Student Paper Award, 2026.
2. (α–β) Zhicheng Du, *Yanwei Sun*, Zihe Wang, Jiahua Wu. Prices as Signals: Joint Design of Information and Pricing. *in preparation for submission to Management Science*.
3. (α–β) *Yanwei Sun*, Jiahua Wu, Jibang Wu. Beyond One Shot: Contracting for Sequential Trials. ***Major Revision at Management Science***.
4. (α–β) Niklas Häusle, Daniel Quigley, Casey Rothschild, *Yanwei Sun*, André Veiga. Optimal Privacy with Coarse Labels in Screening Markets. *in preparation for submission to American Economic Review*.
5. *Yanwei Sun**, Fupeng Sun*, Chiwei Yan, Jiahua Wu. The Role of Prescreening in Auctions with Predictions. ***R&R at Operations Research***.
   - EC 2025 Workshop on Frontiers of Online Advertising.
6. *Yanwei Sun*, Zhe Liu, Chiwei Yan. On-Off Systems with Strategic Customers. ***R&R at Operations Research***.
   - Application with Bay Area Rapid Transit (BART)
7. *Yanwei Sun**, Niloofar Zamani Foroushani*, Zhe Liu, Jiahua Wu. Information Design and Pricing in Lead-Selling Platforms with Supply Competition. ***under review at Manufacturing & Service Operations Management***.
   - Covered by Imperial Business School's IB Knowledge.

Works in Progress

1. Marketplace Design for Agentic Commerce.

2. Pricing AI-model Distillation.

Undergraduate Publications

During my undergraduate study, I was very fortunate to work with Prof. Zhenjun Ming, Prof. Janet K. Allen, and Prof. Farrokh Mistree in the area of ***computational education***.

1. *Yanwei Sun*, Zhenjun Ming, Zachary Ball, Shan Peng, Janet K. Allen, Farrokh Mistree. Assessment of Student Learning Through Reflection on Doing Using the Latent Dirichlet Algorithm. ***Journal of Mechanical Design***, 144(12): 122301, 2022.

Honors and Awards

– Finalist (winner to be announced), Revenue Management and Pricing Jeff McGill Student Paper Award 2026
– Fully Funded Invitation to ISyE–MS&E–IOE Joint Rising Stars Workshop 2025
– First Place, INFORMS Service Science Best Cluster Paper Award 2024
– Second Place, CSAMSE Best Paper Award 2024
– Overseas Research Visit Fund, Imperial Business School (£8,000) 2024
– Imperial College President's PhD Scholarship 2023 – 2027
– Dean's List for Academic Excellence, Imperial Business School 2022 – 2023
– Imperial Business School Fellowship 2022 – 2023

Teaching

Imperial Business School
Teaching assistant for *Data Analytics and Visualisation for Business (MBA)*, Summer 2024
Teaching assistant for *Big Data, AI and Machine Learning* (MBA), Summer 2024
Teaching assistant for *Optimization* (PhD), Fall 2023
Teaching assistant for *Maths and Statistics Foundations for Business Analytics* (MSc), Fall 2023

Academic Service

– Program Committee: *ACM Conference on Economics and Computation (EC), 2026*; *Conference on Web and Internet Economics (WINE), 2025*

– Ad-hoc Award Committee Judge: *INFORMS Public Sector Operations Research Best Paper Award, 2026*

– Ad-hoc Journal Reviewer: *Naval Research Logistics*; *European Journal of Operational Research*

– Conference Reviewer: *ACM Conference on Economics and Computation (EC), 2026*; *Conference on Web and Internet Economics (WINE), 2025*

– Student Co-organizer: *Imperial Analytics & Operations Internal Seminars, 2025 – 2026*

Selected Talks

Pricing AI as a Service

- Revenue Management and Pricing Conference, Ann Arbor, July 2026

Beyond One Shot: Contracting for Sequential Trials

- MSOM Conference, Boston, July 2026
- ACM Conference on Economics and Computation (EC'26), Rome, July 2026

Optimal Privacy with Coarse Labels in Screening Markets

- ACM Conference on Economics and Computation (EC'26), Rome, July 2026

The Role of Prescreening in Auctions with Predictions

- Conference on Web and Internet Economics (WINE), New Brunswick, USA, December 2025
- INFORMS Annual Meeting, Atlanta, October 2025
- Hong Kong University of Science and Technology (IEDA Department), August 2025
- Revenue Management and Pricing Conference, New York, July 2025
- University of Toronto Rotman Young Scholar Seminar Series, virtual, June 2025
- Marketplace Innovation Workshop, May 2025
- University of Chicago (CS Department Seminar), May 2025
- ISyE–MS&E–IOE Joint Rising Stars Workshop, April 2025

Information Design and Pricing in Lead-Selling Platforms with Supply Competition

- INFORMS Annual Meeting, Seattle, WA, October 2024
- ACM Conference on Economics and Computation, Yale School of Management, July 2024
- MSOM Conference, Minneapolis, MN, July 2024

Restricting Entries to All-Pay Contests

- Revenue Management and Pricing Conference, London, July 2023

References

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